

Strength Training for Endurance Runners

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Introduction

Since the 1960s, maximal oxygen uptake ($\dot{V}O_{2\max}$) has been the most popular measurement for assessing performance in distance runners. Early assessments of $\dot{V}O_{2\max}$ recorded values greater than 80 mL kg⁻¹ min⁻¹ in champion athletes [1]. Research has shown strong relationships between $\dot{V}O_{2\max}$ and middle- (800 m, $r = 0.75$) and long-distance (marathon, $r = 0.78$) running performance in heterogeneous groups [2, 3]. However, this relationship is trivial in elite populations (marathon time <2:30, $r = 0.01$) [3]. While a high $\dot{V}O_{2\max}$ (>70 mL kg⁻¹ min⁻¹) may be a pre-requisite to be an elite distance runner, additional physical qualities are needed to succeed at this level. Therefore, other physiological markers such as running economy, fractional utilisation of $\dot{V}O_{2\max}$, velocity at maximal oxygen uptake ($v_{\dot{V}O_{2\max}}$) and sprinting ability play a vital role in world-class performances [4]. This is often demonstrated in middle- and long-distance championship finals, where after a period of high-intensity racing, the winner is generally the runner who can produce the best ‘kick’ during the final lap. For example, during the 2016 Rio Olympic Games 5000 and 10,000 m finals, the gold medallist sprinted at a velocity of over 7.5 m/s during the final stages of each race (see Fig. 23.1).

Strength Training

In addition to a superior cardiovascular system, limitations to world-class distance running performance may therefore be dictated by the neuromuscular system’s rate of force production (RFD). One training technique for improving force production qualities in distance runners is through strength training (see Fig. 23.2). The

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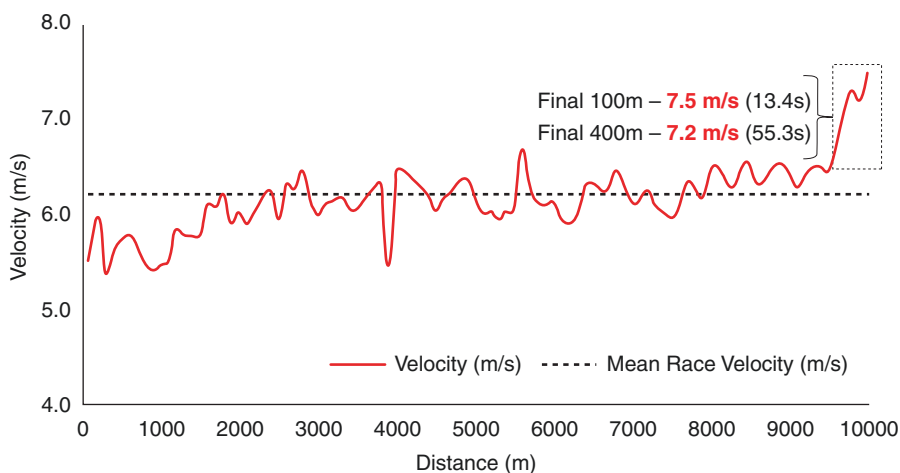


Fig. 23.1 Velocity of the 2016 Rio Olympic Games gold medallist during the 10,000 m final. Note the sudden increase in velocity during the sprint finish (mean race velocity: 6.2 m/s)

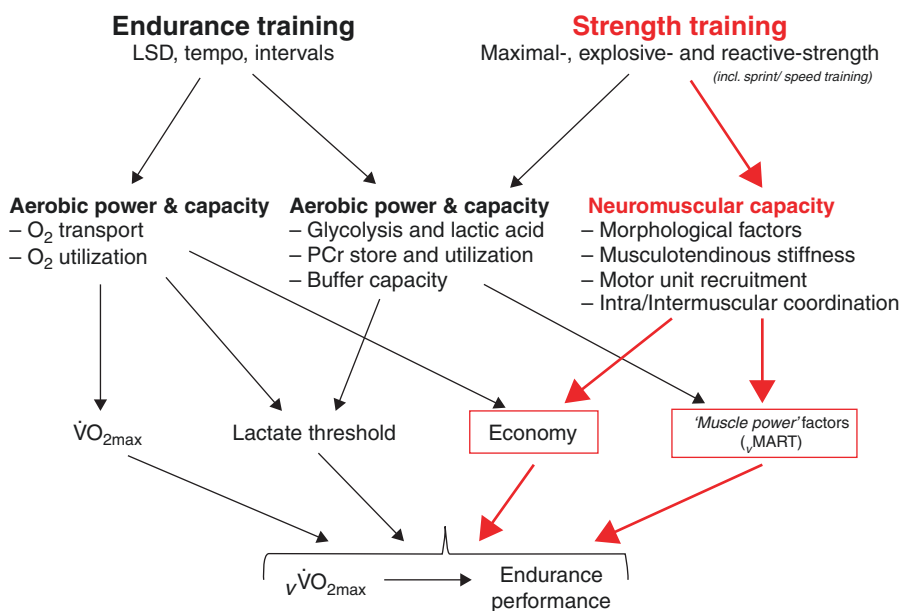


Fig. 23.2 Hypothetical model of the determinants for elite endurance performance and the potential benefits from strength and speed training. Red font and bold arrows highlight the potential benefit of strength and speed training on distance running performance [5] [LSD: long slow distance training; intervals: repeated bouts of exercise lasting 1–8 min and eliciting an oxygen demand equal to 90–100% of $\dot{V}O_{2\max}$; PCr: phosphocreatine; $\dot{V}O_{2\max}$: maximal O_2 uptake; $vMART$: peak velocity in maximal anaerobic running test; $v\dot{V}O_{2\max}$: velocity at $\dot{V}O_{2\max}$]

subcategories for strength training include: (1) maximal-strength (targets maximal force development through high-load, low-velocity movements i.e. squats, trap-bar deadlifts, Bulgarian split-squat); (2) explosive-strength (targets RFD through medium-to-high load, medium-to-high velocity movements i.e. Olympic lifts, jump-squats and squat-jumps); and (3) reactive-strength training (targets musculo-tendinous stiffness and ‘fast’ stretch-shortening cycle function (SSC) through low-load, high-velocity exercises i.e. pogo-jumps, drop-jumps, bounds, sprints). By utilising all three strength methods in an appropriate manner, a ‘strength-trained’ distance runner would theoretically (1) be more economical as the sub-maximal forces produced during each stride would decrease to a lower percentage of maximal force values, and (2) be able to produce higher maximum-velocities through an improved ability to rapidly absorb and create force against the ground.

Research has shown that all types of strength training (maximal-, explosive- and/or reactive-strength) can improve 3 km [6] and 5 km time-trial performance [4], running economy [4, 6–12], $\dot{V}O_{2\max}$ [7, 8, 10] and maximum anaerobic running velocity (v_{MART}) in competitive distance runners [4, 10]. However, it is important to note that most of the strength interventions in previous research studies were relatively short-term (~8 weeks). Additionally, there is a lack of research investigating the effect of sprint training in distance runners. For long-term strength and speed development in distance runners, it is important to prescribe a holistic mixed-methods approach to programming, utilising all strength-qualities (maximal-, explosive- and reactive-strength) and sprint training methods (technical drills, maximum-velocity sprinting) [13, 14]. However, the proportion of each strength quality, and the intensity of sprint training prescribed, will differ depending on the training age of the athlete and the training phase of the macrocycle (i.e. general preparation phase [GPP], special preparation phase [SPP], competition) (see Table 23.4).

Maximum-Strength

Maximal-strength is the ability to voluntarily generate maximum force without a time constraint [15]. Maximal-strength training (also known as the ‘max-effort’ method) targets maximal force development through high-load (>80% one repetition maximum [RM]), low-velocity exercises such as squats and trap-bar deadlifts. The prescription of maximal-strength training has long been a hot topic of debate within the track and field community, with some coaches viewing that maximal-strength training lacks specificity for distance running [7]. However, there has been a growing amount of research demonstrating the effectiveness of maximal-strength training on running economy, $\dot{V}O_{2\max}$ and overall distance running performance [7, 9, 12, 16]. Additionally, maximum-strength training can concurrently improve other strength qualities [17–19] and sprinting ability [20] in relatively weak athletes.

It is important to note that for long-term maximal-strength development, the athlete eventually needs to be exposed to relatively high loads (>80% 1 RM) [21]. High

loads create slow velocities, therefore enabling high forces to be expressed (i.e. the force–velocity relationship). With this in mind, exercise selection may be important. Some track and field coaches contend that for optimal transference, a strength exercise should be ‘specific’ to the running movement (i.e. a low-load, single-leg exercise). However, the aim of maximum-strength training is to promote specific neuromuscular adaptations (e.g. motor-unit recruitment, intramuscular coordination) to increase maximal force capabilities [22]. Consequently, to create an environment where high forces can be expressed, the athlete must be stable. Therefore, traditional multi-joint exercises such as back squats and trap-bar deadlifts provide adequate stability to lift relatively large loads. If the athlete cannot squat or deadlift competently, a single-leg exercise such as a Bulgarian split-squat can provide a stable unilateral alternative. Otherwise, these single-leg exercises can be utilised as an ‘assistance’ exercise following squats or deadlifts to target further strength development (see ‘Assistance Exercises’ section) [23].

A simple linear loading progression provides a sound programming strategy for maximal-strength development in distance runners. During GPP, there may be an initial period where loading is $<80\%$ 1 RM (i.e. 5–8 reps per set). These relatively extensive sets allow technical competency and prepare the musculature for more intensive loading in subsequent training phases. During the latter mesocycles of the GPP, or during the SPP, loading can be increased $>80\%$ 1 RM (i.e. 3–5 reps per set). For distance runners who have an advanced strength training age, partial lifts such as $\frac{1}{2}$ squats and $\frac{1}{4}$ squats can be progressively introduced to increase transfer to explosive-strength and sprinting adaptations [24]. During the competition phase where racing is precedence, concentric or isometric lifts can be incorporated to maintain maximum-strength but reduce likelihood of soreness or fatigue (see Table 23.4).

Maximum-strength can be monitored throughout the macrocycle by RM tests (i.e. 2–5 RM), sub-maximal load-velocity equations or isometric mid-thigh pulls/squats [25]. When the transfer of training diminishes (i.e. lack of improvement in economy, $\dot{V}O_{2\max}$ or maximum-velocity sprinting), the athlete has achieved an adequate level of maximum-strength for distance running. At this point maximum-strength should be maintained, and training should emphasise more specific stimuli such as reactive-strength and sprint training. However, this threshold for diminishing returns may vary between athletes (e.g. $>1.6 \text{ kg} \cdot \text{kg}$ of bodyweight back squat 1RM) and highlights the importance of economy and $\dot{V}O_{2\max}$ monitoring.

Explosive-Strength

Explosive-strength is the ability to produce large forces in minimal time [15]. Explosive-strength training improves RFD and impulse through medium-to-high load, medium-to-high velocity movements (i.e. Olympic lifts, jump-squats, squat-jumps) [15]. However, maximal-strength training is a foundational component of explosive-strength development [21]. Research has shown that the neuromuscular adaptations from maximal-strength training can improve both explosive-strength

and maximal-strength in relatively weak athletes [17–19]. A simple explosive-strength exercise such as the countermovement jump provides a time-efficient exercise prior to (or between sets of) maximal-strength exercises (see Table 23.4). Additionally, the countermovement jump produces the highest maximal power values when compared to other explosive exercises [26]. Regular monitoring of countermovement jump metrics (i.e. jump height, relative peak power), using force platforms, contact mats or phone apps, may help to gain a quick and informative insight into explosive-strength adaptations [25].

Reactive-Strength

Reactive-strength is the ability of the musculotendinous unit to produce a powerful concentric contraction immediately following a rapid eccentric contraction (the stretch-shortening cycle [SSC]) [27]. Research in competitive distance runners has shown that interventions focusing on reactive-strength can improve 3 km [6] and 5 km time-trial performance [4], running economy [4, 6] and v_{MART} [4]. Reactive-strength training is commonly referred to as ‘plyometrics’. Plyometrics, originally coined the ‘shock regime’ by Yuri Verkhoshanky, is a method of jump training that involves a rapid eccentric ‘shock’ stimulation to the musculotendinous unit [28]. The kinematic and kinetic characteristics of fast SSC plyometrics (i.e. bounding, hurdle hops) are similar to that of running (i.e. elastic force production; acute knee and hip angular displacement) [29].

Extensive Plyometrics

Traditional ‘shock’ plyometrics can be demanding on the ankle and knee musculotendinous structures. Therefore, to prepare the athlete for these intensive jumps, it is important to condition the musculotendinous structure during GPP through basic and sub-maximal plyometrics (i.e. pogo-jumps—stiff leg jumps with acute ankle, knee and hip angular displacement). These foundational jumps are also an excellent teaching tool to improve yielding ability (timing of ankle, knee and hip pre-activation prior to ground contact), limb position, coordination and rhythm—characteristics that are required for intensive plyometrics and competent sprinting [30]. Extensive pogo-jumps should be performed in a sub-maximal and rhythmical manner, focusing on dorsi-flexed and flat-footed landings [30]. Pogo-jump height is seldom more than 10 cm, and jump distance between 20 and 50 cm (see Table 23.1).

Intensive Plyometrics

Once the athlete attains an appropriate physiological and technical foundation, jumps can be intensified during the SPP. The [double-leg] pogo-jump from the extensive phase can be intensified by focusing on fast ground contacts and maximal jump height. Due to the increased jump height, there will be an increase in eccentric loading on the landing phase, therefore targeting further SSC adaptations. For distance runners with an intermediate plyometric training age, maximal pogo-jumps can be progressed to tuck jumps (again focusing on fast ground contacts and

Table 23.1 Extensive plyometric suggestions during GPP and SPP for distance runners with a beginner, intermediate, or advanced plyometric training age

Intensity	Sub-maximal (low/moderate)
Frequency per week	2
Distance	2 × 10 m per exercise
Surface	Grass or soft track surface
Reps	20–40 contacts per exercise (i.e. 10–20 contacts per 10 m)
Beginner	Pogo-jumps (double-leg) – 2 × 10 m forwards – 2 × 10 m side-ways – 2 × 10 m backwards
Intermediate/advanced	Pogo-jumps (double-leg) – 2 × 10 m forwards – 2 × 10 m side-ways – 2 × 10 m backwards Pogo-jumps (single-leg) – 2 × 10 m forwards – 2 × 10 m medial side-ways – 2 × 10 m backwards
Cues	<i>‘bouncy’ ‘relaxed’ ‘rhythmical’ ‘land flat-footed’</i>

Table 23.2 Intensive plyometric suggestions during the SPP and competition phases for distance runners with a beginner, intermediate or advanced plyometric training age

Intensity	Maximal (high)
Frequency per week	1–2
Reps	3–8 jumps
Sets	3–5
Recovery	>3 min per set
Beginner	Pogo-jumps
Intermediate	Tuck jumps or hurdle hops
Advanced	Drop-jumps (box height greater than jump height)
Cues	<i>‘get off the ground as quick as possible & jump as high as possible’ ‘bouncy’ ‘land flat-footed’</i>

maximal jump height). Tuck jumps involve ‘tucking’ or flexing the hip and knee during flight (as if the athlete is jumping over hurdles), with a rapid extension prior to landing [29]. Once the athlete is competent at tuck jumping, they can be progressed to hurdle hops. Appropriately spaced hurdles give an excellent external cue to promote jump height while keeping ground contacts low. Lastly, the most intense jumps in the plyometric progression are drop-jumps [28]. Drop-jumps consist of the athlete stepping off a box, landing with minimum ground contact time and jumping for maximum height. Drop-jumps are true ‘shock’ plyometrics due to the supra-maximal eccentric loading (especially when box drop heights exceeds the athlete’s normal vertical jump height) and should only be administered to distance runners with an advanced plyometric training age (see Table 23.2).

It is important to note that throughout the macrocycle (GPP → competition), there should be a fluid and logical progression from extensive to intensive jumps (see Table 23.4). Depending on the athlete's jumping competency and plyometric training age, some elements of the extensive jumps may remain throughout the SPP and competition phases. Reactive-strength is commonly monitored utilising the drop-jump 'reactive-strength index' (RSI). Originally developed at the Australian Institute of Sport in the early 1990s, the drop-jump RSI ($\text{RSI} = \text{jump height} \div \text{contact time}$) is usually assessed from a set box height (i.e. 0.3 m) to control the eccentric stretch loads [31]. Regular monitoring of RSI using force platforms, contact mats or phone apps provide a quick and informative insight into reactive-strength adaptations [25].

Assistance Exercises

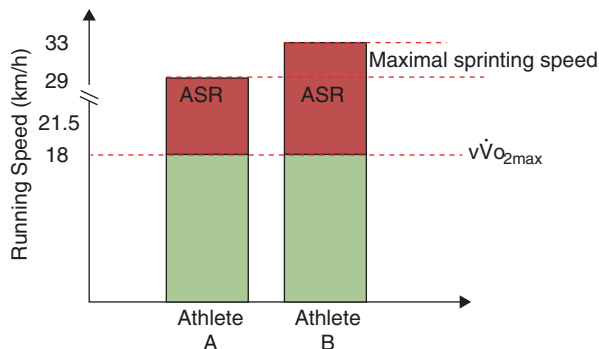
Assistance exercises can be carried out in a circuit format utilising the 'sub-maximal effort' method (8–15 repetition range, 50–80% 1RM) [15]. Depending on the phase of the macrocycle and the physical readiness of the athlete, 1–5 assistance exercises are performed at the end of a strength session. Assistance exercises target the muscles utilised within the 'competition exercise', such as the quadriceps (e.g. split-squat, goblet squat), hamstrings (e.g. short- and long-lever hamstring bridges, Nordics, single-leg RDLs), glutes (e.g. hip thrusts, X-band walks), core (e.g. planks, crunches, lower back) and calves (e.g. straight- and bent-leg calf raises). The aim of assistance exercises is to (1) target further strength development through specific musculature hypertrophy (i.e. quadriceps, hamstring, calves, glutes, abdominals and lower back) [15], (2) improve localised muscular endurance [28] and (3) reduce injury prevalence through improved musculotendinous adaptations and increased pelvic, femoral and overall postural control [32].

Sprint Training

Speed is a vital component of world-class distance running performance. As mentioned earlier, successful world-class distance runners can sprint at velocities over 7.5 m/s during long-distance races, and over 8.5 m/s during middle-distance races [33]. Speed can be enhanced in developmental athletes through general means such as maximal-strength training and plyometrics [20]. However, for optimal speed development, progressive and consistent sprint training is needed [14]. Nevertheless, sprinting is a complex skill. Like most skills, sprinting can be improved through sound technical coaching and appropriate programming. Competent sprint coaching and technical knowledge is common within the track and field sprinting community, however lesser so within distance running.

The neuromuscular adaptations from sprint training (i.e. intermuscular coordination, rate coding, musculotendinous stiffness) may have the potential to improve important physiological factors such as running economy, $\dot{V}\text{O}_{2\text{max}}$ and anaerobic

Fig. 23.3 An illustration highlighting the importance of anaerobic speed reserve (ASR) in distance runners with similar $\dot{V}O_{2\max}$ differing maximal sprinting velocities (adapted from [34])



speed reserve (ASR) [7]. The anaerobic speed reserve (ASR) is the difference between a distance runner's velocity at $\dot{V}O_{2\max}$ ($\dot{V}O_{2\max}$) and their maximal sprinting velocity [34]. Figure 23.3 demonstrates the importance of ASR for two athletes possessing similar $\dot{V}O_{2\max}$, but differing maximal sprinting velocities [34]. During a race (especially middle-distance events where mean race velocities can exceed $\dot{V}O_{2\max}$), an athlete with a large ASR (i.e. Athlete B) will theoretically work at a lower percentage of their maximum-velocity, and will therefore have a lower metabolic load compared to other competitors (i.e. Athlete A). Additionally, a higher maximum-velocity increases an athlete's potential to 'kick' and be successful at the end of a race.

Sprinting Drills

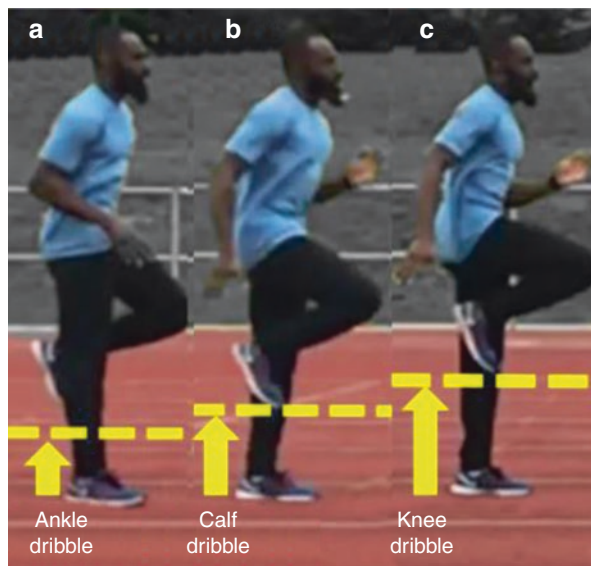
Sprint drills are intended to serve as a simplified or broken-down version of a technically sound sprint cycle [14]. Sprint drills such as dribbles and straight-leg scissors (see below) can be effective for distance runners in rehearsing limb position, rhythm and coordination strategies needed for competent maximum-velocity running. Additionally, these drills provide a specific plyometric conditioning stimulus in areas such as the hamstring and ankle musculotendinous structures.

Dribbles

The mechanics of dribbles are similar to maximum-velocity sprinting; however, the range of motion and velocity of the sprint cycle is truncated. Dribbles reinforce sprint cycle coordination, timing and front-side leg positioning [35]. Additionally, dribbles are less intensive than maximal-velocity sprints, therefore offering an excellent opportunity to practise technically sound sprint mechanics on a regular basis.

There are three types of dribbles in the progression series: ankle-, calf- and knee-dribbles. Teaching progression can start with the ankle-dribble, which has the shortest 'swing-phase' range of motion (see Fig. 23.4a). Once an athlete is competent, they can progress to calf-dribbles (Fig. 23.4b), knee-dribbles (Fig. 23.4c) and finally

Fig. 23.4 The difference in the height of the swing-leg foot during the (a) ankle-dribble, (b) calf-dribble, (c) knee-dribble



dribble-bleeds (blending of a dribble to a full sprint) [35]. Coaching points during a dribble should focus on (1) stepping over the ankle/calf/knee (see Fig. 23.4) (e.g. *‘imagine you are running through knee-high surf/long-grass’*), (2) flat-foot strike directly under centre of mass (see Fig. 23.4), (3) ‘bouncing’ vertically, and (4) circular and symmetrical leg cycles (see Fig. 23.5).

Straight-Leg Scissors

The straight-leg scissor bound drill focuses on the latter part of the late-swing phase of a sprint cycle, emphasising the long-lever ‘down-strike’, pre-activation of posterior leg musculature prior to ground contact, and hip projection during the stance phase. Additionally, this drill provides a specific plyometric stimulus for conditioning the hamstring musculature at long lengths. Coaching points during a straight-leg scissor bound should focus on (1) straight leg, (2) ‘snapping’ the leg/foot downwards and backwards, and (3) ‘bouncing’ vertically and horizontally (i.e. projecting the hip vertically and horizontally) (Fig. 23.6).

Warm-Ups and ‘Bleed’ Drills

Dribbles and scissor drills are best utilised during the warm-ups of sprint and strength sessions, endurance sessions and races (see Table 23.4). Towards the end of the warm-up, ‘bleed’ drills can be incorporated to blend the dribble or scissor drill into the full sprint cycle. For example, a dribble-bleed would start off as a knee-dribble (see Fig. 23.5c), however once the athlete is dribbling at a high speed (i.e. around 20–30 m), the knee-dribble is smoothly ‘bled’ into a full sprint cycle. The kinaesthetic feeling of rhythmically stepping over the knee and bouncing off the ground during a knee-dribble is instantaneously incorporated into a full sprint cycle.

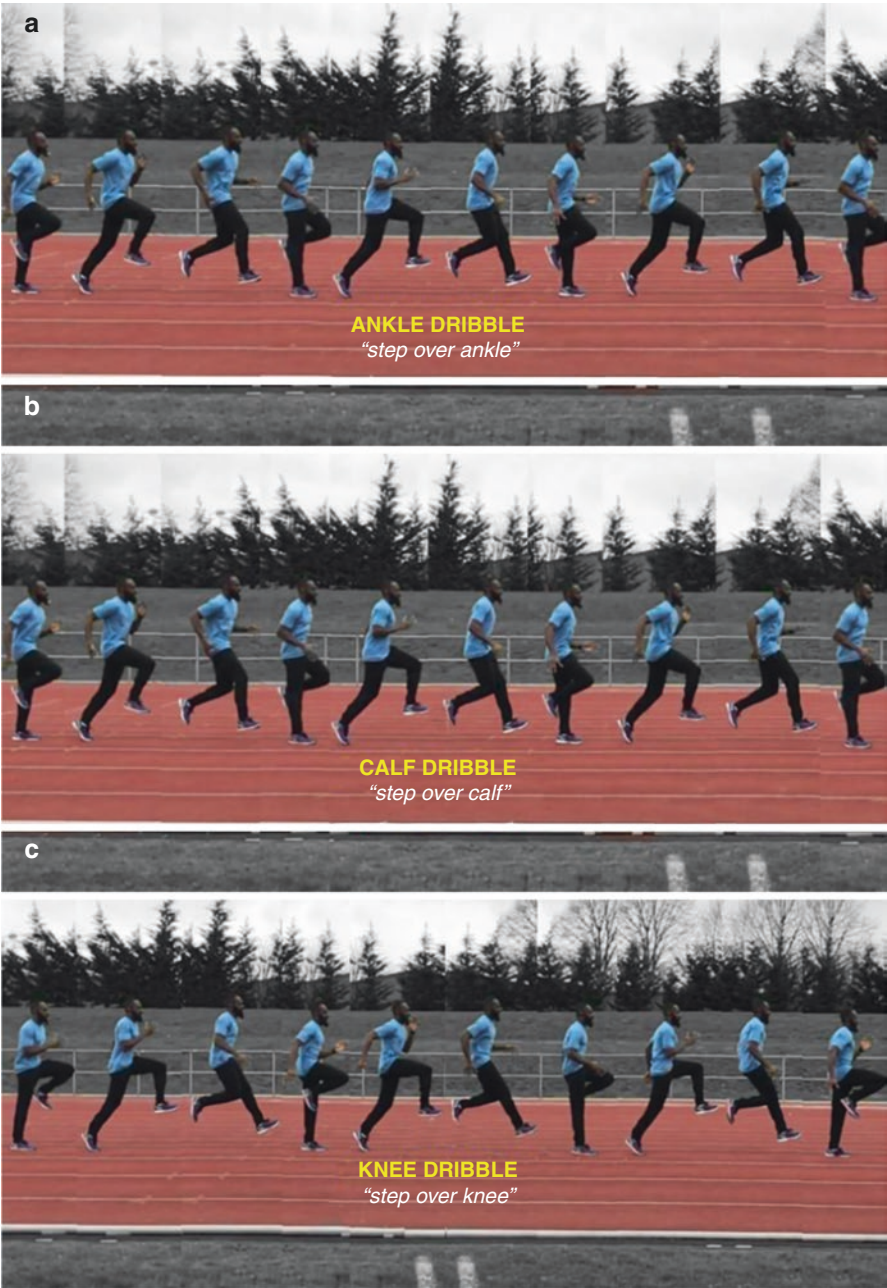


Fig. 23.5 The circular and symmetrical leg cycle during the (a) ankle-dribble, (b) calf-dribble, (c) knee-dribble



Fig. 23.6 The leg cycle during a straight-leg scissor bound

Table 23.3 Sprint training and warm-up suggestions for distance runners during GPP, SPP and competition

Intensity	Maximal (high)
Frequency per week	1–2
Reps	5–10
Recovery	3–5 min
GPP	5–10 × [10–20 m accelerations or sub-max flying runs]
SPP/competition	5 × [30–40 m sprints or flying runs]
Warm-up	2 × 20 m: – Ankle-dribble – Calf-dribble – Knee-dribble – Straight-leg scissors – Dribble-bleed – Scissor-bleed

Sprint Training

To prepare distance runners for running at maximal velocities, sprinting should be introduced in a gradual and progressive manner. One approach to speed training is to initially focus on short, maximal accelerations (from a standing start) during the GPP phase (i.e. 10–20 m), and progress by building sprint distances every micro-/mesocycle until maximum-velocity is attained (i.e. 30 or 40 m). Another approach is to perform ‘flying sprints’ from a rolling start, with a longer sub-maximal (and less intensive) acceleration. This type of run-up allows maximum-velocity to be reached without excessive anaerobic cost and fatigue. Once maximum-velocity is attained, it is held for a set distance (i.e. 10 m). This session can be introduced in a progressive manner during GPP, where the athlete initially focuses on accelerating to 80% of maximum, and progressing their velocity every mesocycle (see Table 23.4). Once the athlete is conditioned for sprinting at maximum-velocity for 10 m, this distance can be gradually progressed (i.e. 20–30 m). To ensure speed development, it is important that each repetition consists of maximal effort, with full recoveries in between repetitions (5–10 repetitions, with 3–5 min recovery between). Sprint development requires quality, not quantity (see Table 23.3).

Programming

There has been a growth in the literature investigating the compatibility of strength and aerobic training (i.e. concurrent training) and the effect on the mechanisms underpinning protein synthesis [36, 37]. Therefore, it is important that coaches are aware of this potential ‘interference effect’ associated with combined strength and endurance training. While it has been demonstrated that high volume endurance training may compromise the absolute gains in maximal-strength [38], the neuromuscular gains achieved through regular strength training in endurance athletes have been shown to be sufficient for improving endurance performance. However, for optimal adaptation and development of a distance runner, speed and strength sessions (both combined together, see Table 23.4) should be carefully programmed around ‘intense’ aerobic endurance training (i.e. zone 4 and 5, >LT2 > 80% $\dot{V}O_{2max}$) —possibly on the same days as recovery sessions or low-intensity aerobic training. Research has found that successful elite distance runners spend approximately 80% of their training in these low-intensity, aerobic-dominant training zones [39], therefore giving opportunities to programme strength and speed training sessions without hampering the preparation or recovery of more specific and intense aerobic sessions.

For long-term neuromuscular development in distance runners, it is advised to prescribe a mixed-methods approach to programming, utilising all strength-qualities (maximal-, explosive- and reactive-strength) and sprint training methods (technical drills, maximum-velocity sprinting) within the same session [13, 14]. However, the proportion and volume of each strength quality (including speed training) will differ depending on the training age of the athlete and the training phase of the macrocycle (i.e. general preparation phase [GPP], special preparation phase [SPP], competition)

Table 23.4 Strength and speed programming design during GPP, SPP and competition

	GPP	SPP	Competition
Warm-Up (Sprint Drills)	Sprint Drills (i.e dribbles, straight-leg scissors, bleeds)		
1. Speed	Sub-max sprints (i.e 80% of max V)	10m 'Flying' sprints (i.e 90% of max V)	10m 'Flying' sprints (i.e 100% of max V)
2. Reactive-Strength	Extensive Plyometrics (i.e sub-max pogo-jumps)	Ext / Int Plyometrics (i.e maximal pogo-jumps)	Intensive Plyometrics (i.e tuck jumps)
3. Explosive-Strength	Countermovement Jumps (i.e 3 × 3 – can be implemented between sets of squats)		
4. Maximal-Strength	Squats (full ROM)* (i.e 3 × 8)	½ Squat* (i.e 3 × 5)	¼ Squat (Con / Iso)* (i.e 3 × 3)
5. Assistance	Single-leg squats, hamstrings, glutes, core, calves (i.e 1 – 5 exercises 'circuit' style, 8 – 15 repetitions)		

max V maximum-velocity, *ext* extensive, *int* intensive, *ROM* range of motion, *con* concentric, *iso* isometric

^aCan be replaced with trap-bar deadlift or a single-leg squat alternative

(see Table 23.4). When programmed and coached appropriately, the combined speed and strength session (including sprint drills warm-up) should last no more than 45–60 min. For full-time distance runners, it is advised that the combined speed and strength session should be prescribed 1–2 times per week during the GPP and SPP, and once every 7–10 days during competition phases [7]. However, for experienced distance runners who are already ‘strong’ and have high force capabilities, there may need to be greater emphasis on intensive plyometrics and maximal-velocity sprinting to gain further improvements in speed, economy and $\dot{V}O_{2\max}$.

Summary

Exercise economy, $\dot{V}O_{2\max}$ and sprinting ability play a vital role in elite distance running performance. Research has shown that the neuromuscular adaptations from strength training can improve these key performance indicators. Therefore, strength and sprint training can be useful for improving elite distance running performance. For long-term strength and speed development, it is important to prescribe a mixed-methods approach to strength (maximal-, explosive- and reactive-strength) and speed programming (technical drills and maximum-velocity training). However, the focus of each strength quality, and the intensity of speed training prescribed, will depend on the phase of the macrocycle (i.e. GPP, SPP, competition) and the training age of the athlete. Additionally, for optimal adaptations, strength and speed sessions should be carefully programmed around ‘intense’ aerobic training and racing.

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